

Phylogenetic inference and likelihood

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(With thanks to Mark Holder and Paul Lewis for slides)

This pattern is surprising.

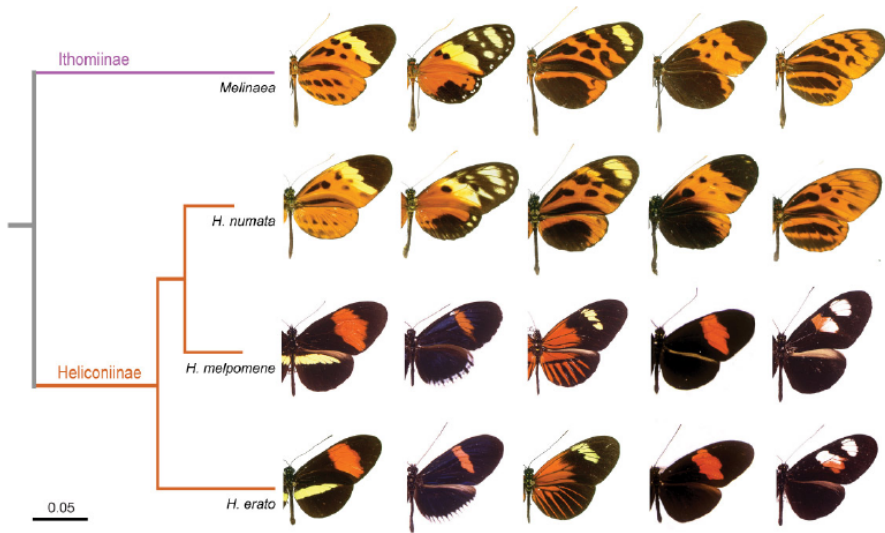
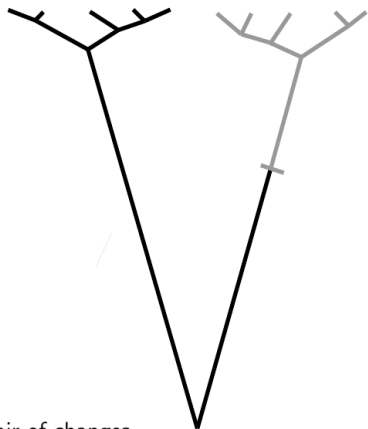


Figure by Mathieu Joron: <http://xyala.cap.ed.ac.uk/joron/>

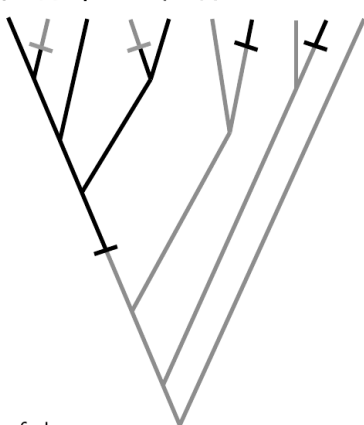
These two trees tell very different evolutionary stories.

X	X	X	X	X	-	-	-	-	-
T	T	T	T	T	A	A	A	A	A



1 pair of changes.
Coincidence?

X	-	X	-	X	-	X	-	X	-
T	A	T	A	T	A	T	A	T	A



5 pairs of changes.
Much more convincing

How do we figure out what tree captures the relationships we're interested in?

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 - ▶ Current patterns of biodiversity has been generated by processes of: (1) speciation, (2) extinction, and (3) character modification.

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- ▶ Fundamental perspectives in all these approaches:
 - ▶ Current patterns of biodiversity has been generated by processes of: (1) speciation, (2) extinction, and (3) character modification.
 - ▶ The phylogeny is an abstract representation (“model”) of this diversification process.

Enormous numbers of topologies to consider

<u>Taxa</u>	<u>Unrooted binary trees</u>	<u>Rooted binary trees</u>
3	1	3
4	3	15
5	15	105
6	105	945
7	945	10,395
8	10,395	135,135
9	135,135	2,027,025
10	2,027,025	3×10^7
15	7×10^{12}	2×10^{14}
20	2×10^{20}	8×10^{21}
50	3×10^{74}	
100	2×10^{182}	
1,000	2×10^{2860}	
10,000	8×10^{38658}	
1,000,000	1×10^{5866723}	

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it is estimated that there are between 10^{78} to 10^{82} atoms in the known, observable universe.

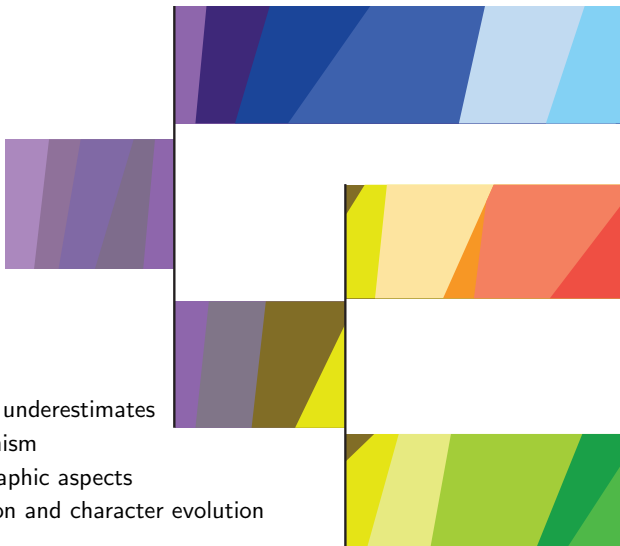
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Estimating a tree from character data

Tree construction:

- ▶ strictly algorithmic approaches - use a “recipe” to construct a tree
- ▶ optimality based approaches - choose a way to “score” a trees and then search for the tree that has the best score.

Phylogeny with complete genome + “phenome” as colors:



This figure:
dramatically underestimates
polymorphism
ignore geographic aspects
of speciation and character evolution

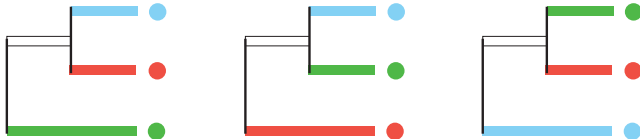
Extant species are just a thin slice of the phylogeny:

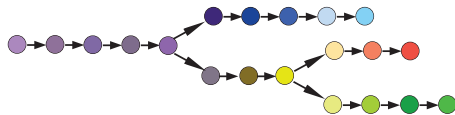


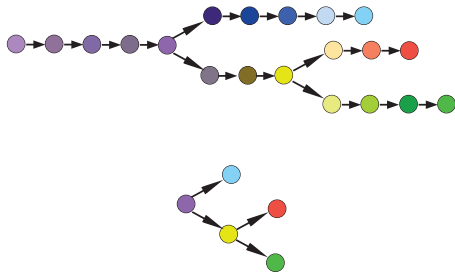
Our exemplar specimens are a subset of the current diversity:

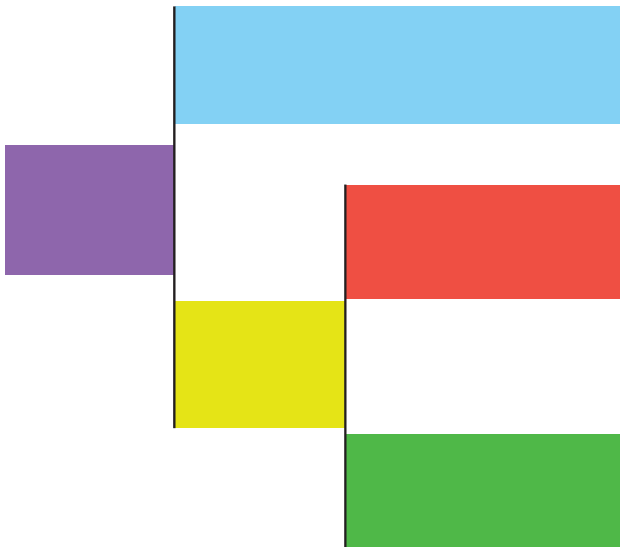
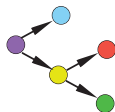


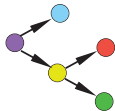
The phylogenetic inference problem:











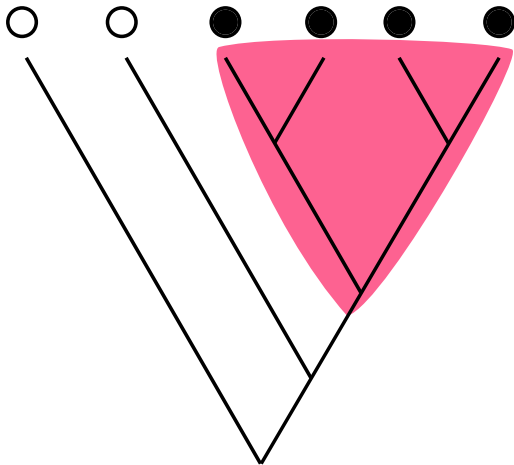
Multiple origins
of the yellow state
violates our assumption
that the state codes in
our transformation scheme
represent homologous states

The meaning of homology (**very roughly**):

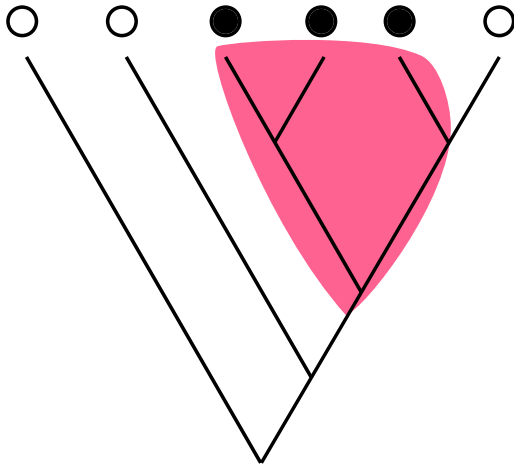
1. comparable (when applied to characters)
2. identical by descent (when applied to character states)

Ideally, each possible character state would arise once in the entire history of life on earth.

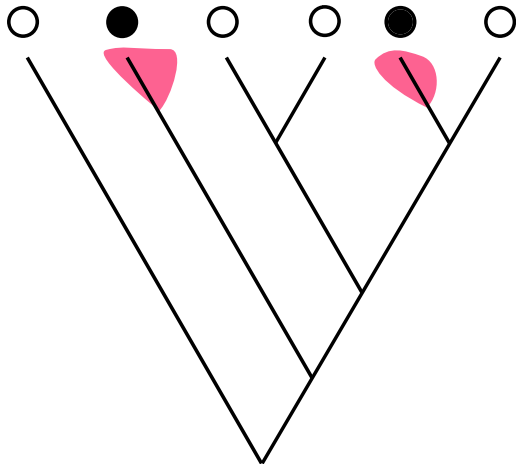
Instances of the filled character state are homologous
Instances of the hollow character state are homologous



Instances of the filled character state are homologous
Instances of the hollow character state are NOT homologous



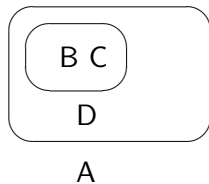
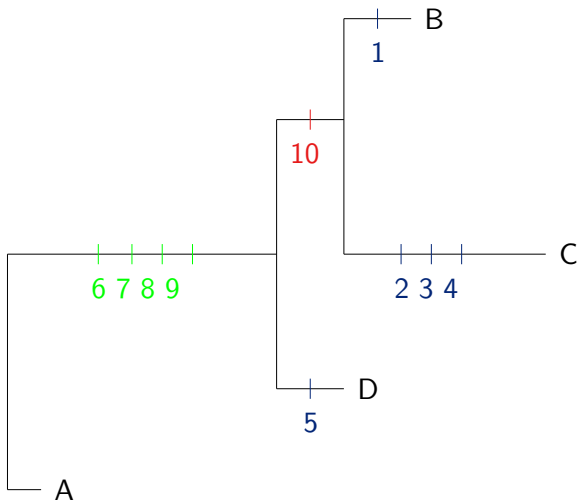
Instances of the filled character state are NOT homologous
Instances of the hollow character state are homologous



If 0 is the ancestral state, and 1 is the derived state, and there is no homoplasy, what are the relationships between these taxa?

Taxon	Character #									
	1	2	3	4	5	6	7	8	9	10
A	0	0	0	0	0	0	0	0	0	0
B	1	0	0	0	0	1	1	1	1	1
C	0	1	1	1	0	1	1	1	1	1
D	0	0	0	0	1	1	1	1	1	0

Don't look at the next slide yet!!



If characters are not polarized (ancestral and descendent states known) can infer unrooted trees.

Character conflict

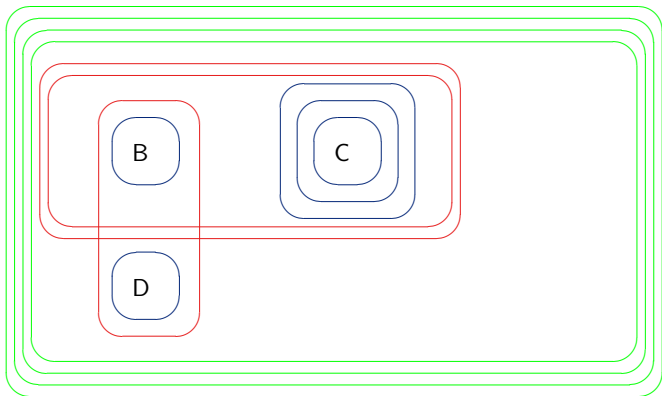
<i>Homo sapiens</i>	A G TTCAAG T
<i>Rana catesbiana</i>	A A TTCAAG T
<i>Drosophila melanogaster</i>	A G TTCAAG C
<i>C. elegans</i>	A A TTCAAG C

The red character implies that either (*Homo* + *Drosophila*) is a group (if G is derived) and/or (*Rana* + *C. elegans*) is a group.

The green character implies that either (*Homo* + *Rana*) is a group (if T is derived) and/or (*Drosophila* + *C. elegans*) is a group.

The green and red character cannot both be correct.

Taxon	Character #											
	1	2	3	4	5	6	7	8	9	10	11	12
A	0	0	0	0	0	0	0	0	0	0	0	0
B	1	0	0	0	0	1	1	1	1	1	1	1
C	0	1	1	1	0	1	1	1	1	1	1	0
D	0	0	0	0	1	1	1	1	1	0	0	1



A

Character conflict

Two characters are compatible if they can both be mapped on the same tree so that all of the character states displayed could be homologous.

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Incompatible characters are evidence of homoplasy in the data

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Incompatible characters are evidence of homoplasy in the data

Homoplasy literally means the “same change” has occurred more than once in the evolutionary history of the group. The presence of homoplasy undermines Parsimony analyses.

How do we estimate phylogenies in the presence of character conflict?

We will focus on Maximum Likelihood and Bayesian statistical estimates for evolutionary models.

When should we expect character conflict?

- ▶ Data type?
- ▶ Evolutionary history?

How can we deal with character conflict?

- ▶ We have the data! We are trying to figure out how it was generated.
- ▶ Likelihood provides a measure of surprise under different models

What is the probability of getting a 2 heads for two coin flips under each of these models:

- ▶ Fair coin
- ▶ Two heads
- ▶ Two tails

How likely of each of these models given that you got 2 heads:

- ▶ Fair coin
- ▶ Two heads
- ▶ Two tails

Does that mean that this coin likely has 2 heads?

How likely of each of these models given that you got 2 heads:

- ▶ Fair coin
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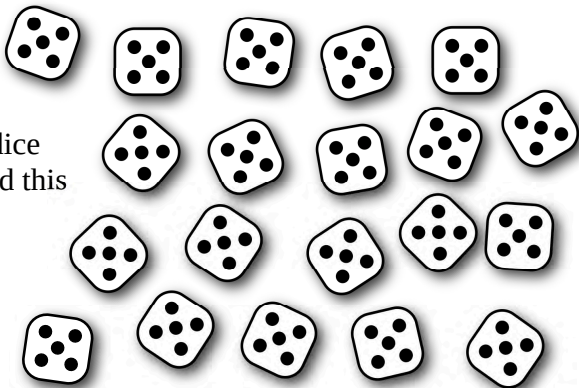
In a Bayesian approach you would incorporate data about your prior expectation that each of these models was correct.

The Likelihood Criterion

The probability of the observations computed using a model tells us how surprised we should be.

The preferred model is the one that surprises us least.

Suppose I threw 20 dice down on the table and this was the result...



Combining probabilities

- *Multiply* probabilities if the component events must happen **simultaneously** (i.e. where you would naturally use the word AND when describing the problem)

Using 2 dice, what is the probability of

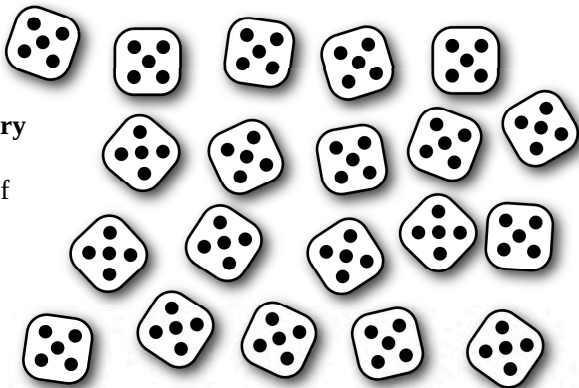


$$(1/6) \times (1/6) = 1/36$$

The Fair Dice model

$$\Pr(\text{obs.} | \text{fair dice model}) = \left(\frac{1}{6}\right)^{20} = \frac{1}{3,656,158,440,062,976}$$

You should have been **very surprised** at this result because the probability of this event is **very small**: only 1 in 3.6 quadrillion!

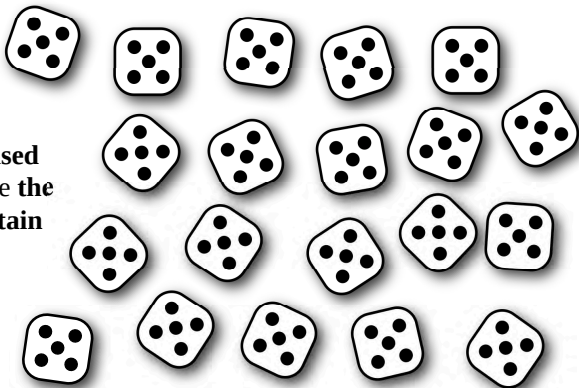


The Trick Dice model

(assumes dice each have 5 on every side)

$$\Pr(\text{obs.} | \text{trick dice model}) = 1^{20} = 1$$

You should **not be surprised at all** at this result because **the observed outcome is certain** under this model



Results

Model	Likelihood	Surprise level
Fair Dice	$\frac{1}{3,656,158,440,062,976}$	Very, very, very surprised
Trick Dice	1	Not surprised at all

winning model maximizes likelihood
(and thus minimizes surprise)

Likelihood: why a new term?

Outcome	Fair coin model	Two-heads model
H	0.5	1
T	0.5	0
	1	1

Likelihoods of models given one particular data outcome are *not* expected to sum to 1.0

Probabilities of data outcomes given one particular model sum to 1.0

Maximum Likelihood phylogenetics

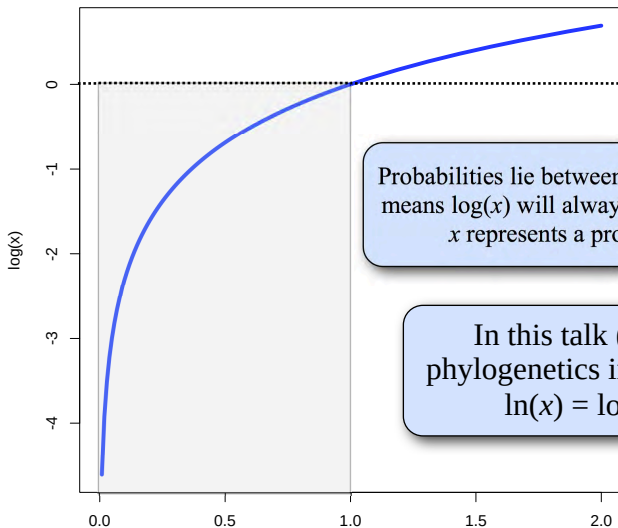
- ▶ What tree makes our data the least surprising?
- ▶ Answering this question requires answering a lot of other questions too!
 - ▶ What do we expect our data to look like under different scenarios?
 - ▶ What is our model of evolution?

Likelihood and model comparison

- Analyses using likelihoods ultimately involve **model comparison**
- The models compared can be **discrete** (as in the fair vs. trick dice example)
- More often the models compared differ **continuously**:
 - Model 1: branch length is 0.01
 - Model 2: branch length is 0.02
 - Model 3: branch length is 0.03

Rather than having an infinity of models, we instead think of the branch length as a **parameter** within one model

Likelihoods vs. log-likelihoods



Probabilities lie between 0 and 1, which means $\log(x)$ will always be negative if x represents a probability.

In this talk (and in phylogenetics in general),
 $\ln(x) = \log(x)$

Likelihood calculated from a single sequence

$$\Pr(A) = \pi_A$$

$$\Pr(C) = \pi_C$$

$$\Pr(G) = \pi_G$$

$$\Pr(T) = \pi_T$$

First 32 nucleotides of the $\psi\eta$ -globin gene of gorilla:

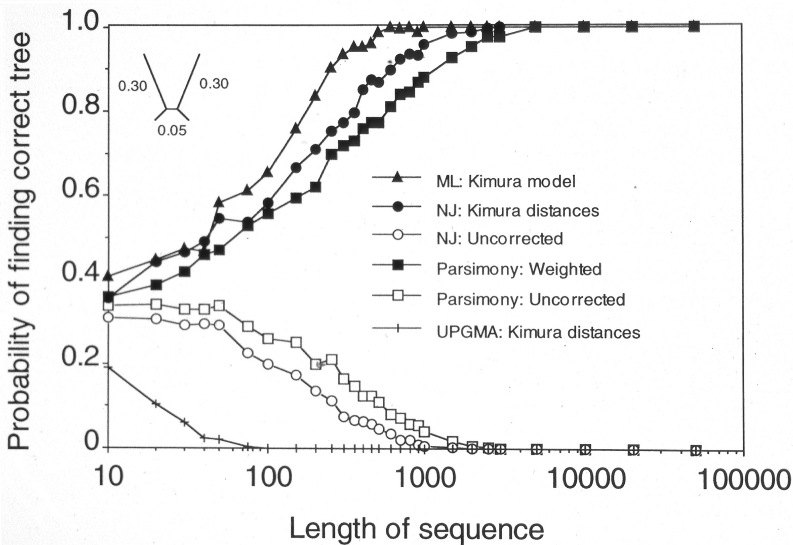
GAAGTCCTTGAGAAATAAACTGCACACACTGG

$$\begin{aligned} L &= \pi_G \pi_A \pi_A \pi_G \pi_T \pi_C \pi_C \pi_T \pi_T \pi_G \pi_A \pi_G \pi_A \pi_A \pi_A \pi_T \pi_A \pi_A \pi_A \pi_C \pi_T \pi_G \pi_C \pi_A \pi_C \pi_A \pi_C \pi_A \pi_C \pi_T \pi_G \pi_G \\ &= \pi_A^{12} \pi_C^7 \pi_G^7 \pi_T^6 \end{aligned}$$

Note that we are assuming independence among sites here

$$\log L = 12 \log(\pi_A) + 7 \log(\pi_C) + 7 \log(\pi_G) + 6 \log(\pi_T)$$

We can already see by eye-balling this that a model allowing **unequal** base frequencies will **fit better** than a model that assumes **equal** base frequencies because there are about twice as many As as there are Cs, Gs and Ts.

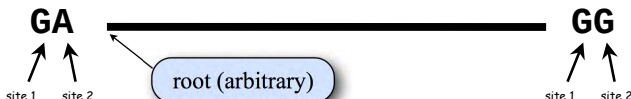


Hillis, D. M., J. P. Huelsenbeck, and D. L. Swofford. 1994. Hobgoblin of Phylogenetics? Nature 369:363-364.

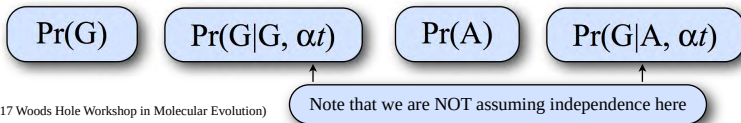
Likelihood of the simplest tree

sequence 1 sequence 2

To keep things simple, assume that the sequences are only 2 nucleotides long:



$$\begin{aligned}
 L &= L_1 L_2 \\
 &= \left[\begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad \left(\frac{1}{4} + \frac{3}{4}e^{-4\alpha t} \right) \right] \left[\begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad \left(\frac{1}{4} - \frac{1}{4}e^{-4\alpha t} \right) \right]
 \end{aligned}$$



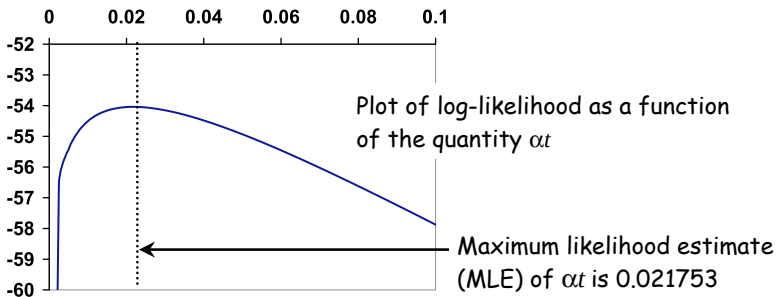
Maximum likelihood estimation

First 32 nucleotides of the $\psi\eta$ -globin gene of gorilla and orangutan:

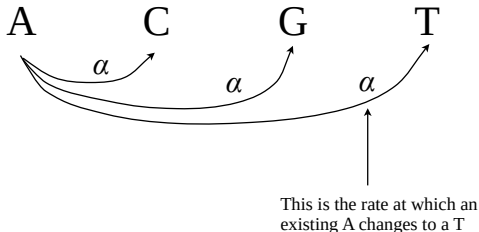
gorilla **GAAG**TCCTTGAGAAATAAACTGCACACACTGG

orangutan **GAATC**TCCTTGAGAAATAAACTGCACACACTGG

$$L = \left[\left(\frac{1}{4} \right) \left(\frac{1}{4} + \frac{3}{4} e^{-4\alpha t} \right) \right]^{30} \left[\left(\frac{1}{4} \right) \left(\frac{1}{4} - \frac{1}{4} e^{-4\alpha t} \right) \right]^2$$



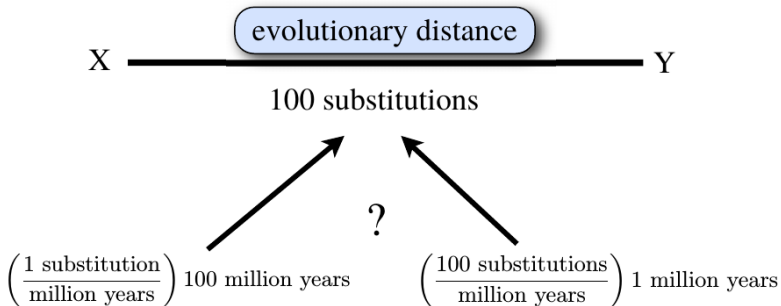
number of substitutions = rate $\times$ time



Overall substitution rate is 3α , so the expected number of substitutions (v) is

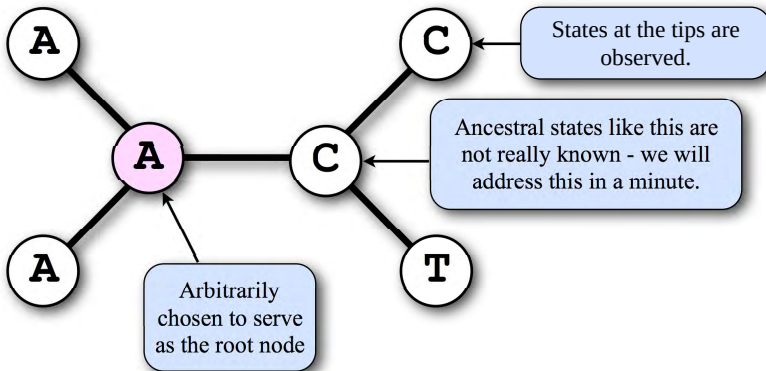
$$v = 3\alpha t$$

Rate and time are confounded

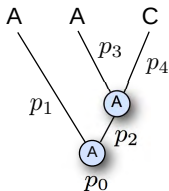


Likelihood of an unrooted tree

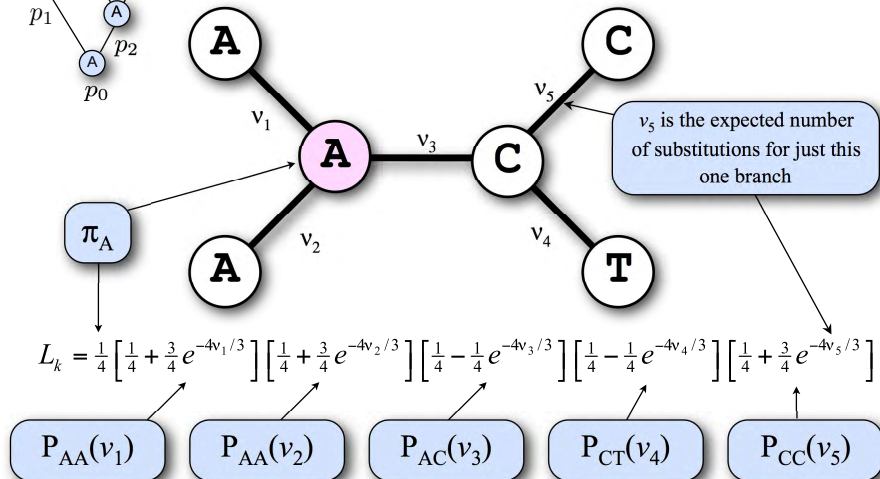
(data shown for only one site)



From slide 6



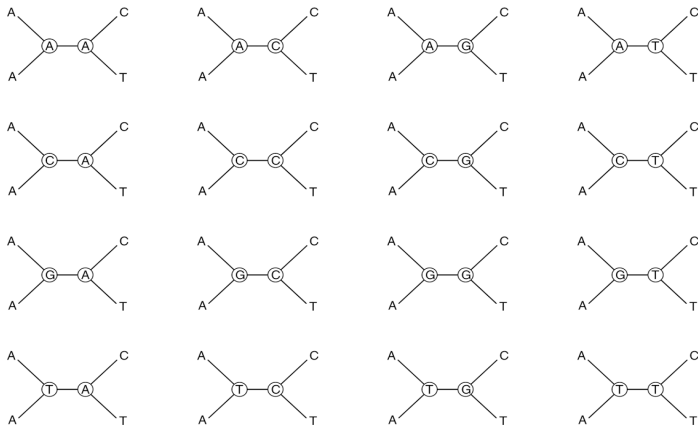
Likelihood for site k



How do we know the states at the internal nodes?

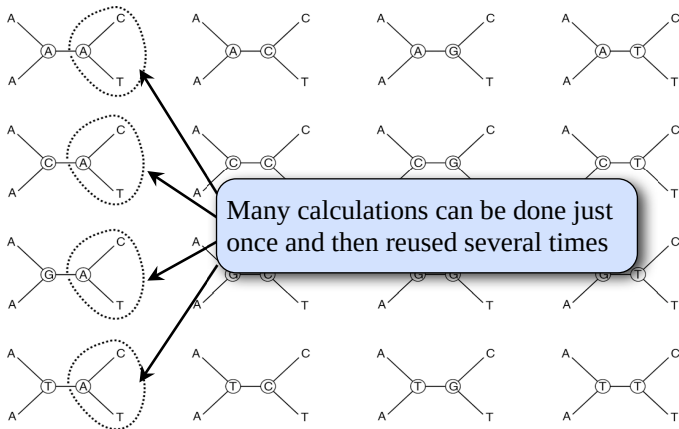
How do we know the states at the internal nodes?
We don't! We must consider all possible paths.

Brute force approach would be to calculate L_k for all 16 combinations of ancestral states and sum them



Note use of the OR probability rule

Pruning algorithm (same result, less time)



Felsenstein, J. 1981. Evolutionary trees from DNA sequences:
a maximum likelihood approach. *Journal of Molecular Evolution* 17:368-376

We will explore likelihood of different trees using these super cool widgets developed by Mark Holder:

- ▶ <http://phylo.bio.ku.edu/mephytis/barcharts.html>
- ▶ <http://phylo.bio.ku.edu/mephytis/brlen-opt.html>
- ▶ <http://phylo.bio.ku.edu/mephytis/tree-opt.html>